Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/24/2019
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Project Title:	Cyclonic Two-Phase Flow Separator Experiment in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan Ph.D. (Case Western Reserve University)		
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Task Description:	NOTE that this is a Flight Experiment follow-on to "Investigation into the Quantification and Control of Steady and Unsteady Flow Phenomena in Passive Gas-Liquid Cyclonic Separators" grant NNX09AI31G with same principal investigator. Multiphase flow separation is a critical technological challenge for the human exploration of space and planetary bodies. While droplet and bubble segregation is a spontaneously occurring phenomenon in most terrestrial situations, the absence of the gravity-induced separatory action in a microgravity environment often results in situations where two disparate phases have no distinct inclination to separate from one another. This can present many problems with respect to spacecraft and space station operations as proper control and sequestration of the individual phases is often essential to the successful functionality of many system-critical components. Passive cyclonic separators are a promising technology that potentially could fulfill the separation needs of many of these systems. While considerable research has been performed on this technology, the limitations of terrestrial testing, aircraft testing, and modeling necessitates long-term experimentation to raise these devices to a sufficient technology readiness level (TRL) to permit use within spacecraft systems. It is a lack of basic engineering and scientific knowledge that currently prevents these systems from being employed. At a basic engineering level, there is a distinct lack of data regarding both the steady and unsteady operational performance of vortex separators. Issues such as where the designs can be expected to work and how to go about optimizing designs are unresolved. Accurate prediction of these phenomena using computational fluid dynamics (CFD) is currently not tenable due to the dearth of available data and incomplete modeling techniques. Beyond engineering, there is also a lack of knowledge regarding the fundamental multiphase fluid physics phenomena that occur within passive cyclonic separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. are all unresolved. Inasmuch, this research effort seeks to resolve these issues. The main objectives are to: 1. Quantitatively assess and delimit the range of operability of a prototype passive cyclonic separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct. 2. Elucidate and quantify the multiphase fluids physics phenomena governing the performance of microgravity two-phase separators. 3. Validate and verify the multiphase CFD modeling approaches that have been developed to simulate the complicated two-phase transport processes associated with phase separation in microgravity. 4. Validate and verify a hydraulic system code for the simulation of two-phase flow in microgravity. To achieve these objectives, combined experimental, numerical, and phenomenological modeling analyses are being performed to quantitatively describe the performance and response of systems containing separators. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities. The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a microgravity/scientific standpoint, the experimentation, modeling, and analytical work of this research will serve to: 1. Provide elucidation and understanding of the above-mentioned two-phase flow phenomena associated with the operation of separators in multiphase systems 2. Provide details of the columnar stability of the gas core 3. Provide details of the time constants associated with the response of passive cyclonic separators 4. Validate and verify a coupled CFD and hydraulic system code for simulations of two-phase flow in microgravity The resulting insight into phase separation, distribution, and control will help to afford NASA designers the latitude to take greater advantage of the benefits offered by the use of these multiphase systems in spacecraft applications. The experimental results will also facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new separator designs, while the results from the system code are expected to assist in the creation and analysis of general multiphase systems in microgravity. Overall, the research will help contribute to the fundamental scientific understanding of two-phase fluid behavior, both in microgravity and 1-g environments.
	Numerical Modeling Our work on the collapse of the gas core due to surface tension was reported in the last year’s report. In order to further study the gas core stability and bubble dynamics, a 3D separator model is required for CFD simulation. The 3D computational domain covers the region above the baffle plate including an injection nozzle, because this region is of our primary interest. Two-phase mixture enters the computational domain through the nozzle with prescribed velocities for both phases or with prescribed volume fraction. The gas exits the domain through a gas outlet pipe whose end is maintained at atmospheric pressure. The gap between the baffle plate and the separator cylinder wall is modeled as the liquid outlet. At the liquid outlet, the pressure is set equal to the overall pressure drop in the liquid line beyond the outlet. The pressure drop is proportional to the square of the liquid flow rate according to earlier experimental results. The no-slip boundary condition is applied at all walls. The computational grid of the 3D separator model is constructed with structured mesh which contains approximately 400,000 hexahedrons.

Task Progress:	The gas core radii for various injection Reynolds numbers have been computed using the 3D model. The results are compared with earlier 2D simulation results as well as with our experimental data. Also, the velocity fields in the liquid film are compared with the 2D simulations. Both comparisons of gas core radius and velocity profiles show reasonable agreement, which validates the 3D model. Analytical Modeling In our earlier work, a control volume analysis had been performed and gave reasonably accurate results. However, the angular momentum going out of the separator was overestimated based on the assumption that the liquid annulus is moving in a uniform solid body rotation. Furthermore, the effect of a scaling parameter G, which represents the ratio of cross-sectional area of the separator body to the injection nozzle, was found to be important and must be included in the expression of the friction coefficient C_f. Thus, in the current study, the control volume equations have been re-derived. Although the present research only focuses on liquid-only injections, the system of equations can be further extended to two-phase injection. With combined analytical and numerical techniques, a new system of control volume equations is derived, which can both qualitatively and quantitatively describe the key features of the separator. The equations were used to predict the gas core size formed in the separator under various injection Reynolds numbers and nozzle sizes. The predicted results agree reasonably well with the experimental data; thus, the control volume equations can be used as a guidance for future design of the device.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Abstracts for Journals and Proceedings	Li Y, Kamotani Y. "Numerical analysis of gas-liquid two-phase flow separator performance in microgravity." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018